

Math 290

Practice Quiz 2 — Version F Solutions

1(a) Answer: D. The step “ $|x - a| = x - a$ ” is only valid when $x - a \geq 0$. The missing case: if $x - a < 0$, then $|x - a| = a - x$, so $0 < a - x \leq b$; multiplying by -1 gives $-b \leq x - a < 0$, and adding a gives $a - b \leq x < a \leq a + b$. (Based on Proposition 8.20.)

1(b) Answer: False. The negation of $\exists x \in \mathbb{R}, x^2 < -1$ is the *universal* statement $\forall x \in \mathbb{R}, x^2 \geq -1$, so a disproof must show this for every x : let $x \in \mathbb{R}$; then $x^2 \geq 0 > -1$. One example is not enough. (Based on Section 11.D.)

1(c) Answer: True. $(28, 6) \in \mathbb{Z} \times \mathbb{Z}$ and $28 - 6 = 22 = 11 \cdot 2$, so $28 \equiv 6 \pmod{11}$. (Based on Proposition 10.2.)

2. (Based on Exercise 7.8.)

Solution. “ P only if Q ” means $P \Rightarrow Q$, so we must prove: if $5x - 1$ is even, then x is odd.

Directly: Assume $5x - 1$ is even, so $5x - 1 = 2k$ for some $k \in \mathbb{Z}$. Then $x = (5x - 1) - 4x + 1 = 2k - 4x + 1 = 2(k - 2x) + 1$, which is odd since $k - 2x \in \mathbb{Z}$.

Contrapositively (also fine): Assume x is even, $x = 2m$. Then $5x - 1 = 10m - 1 = 2(5m - 1) + 1$ is odd. □

3. (Based on Proposition 11.10 and the uniqueness technique of Proposition 11.11.)

Solution. Let $n \in \mathbb{Z}$.

(Existence) There are two cases. Case 1: n is even, say $n = 2k$ with $k \in \mathbb{Z}$. Fix $m = k$. Then $2m = n < n + 2 = 2m + 2$. Case 2: n is odd, say $n = 2k + 1$ with $k \in \mathbb{Z}$. Fix $m = k$. Then $2m = 2k < 2k + 1 = n < 2k + 2 = 2m + 2$. In both cases $2m \leq n < 2m + 2$.

(Uniqueness) Assume $m, m' \in \mathbb{Z}$ both satisfy $2m \leq n < 2m + 2$ and $2m' \leq n < 2m' + 2$. Then $2m \leq n < 2m' + 2$, so $m < m' + 1$; since m and m' are integers, $m \leq m'$. By the symmetric argument ($2m' \leq n < 2m + 2$), $m' \leq m$. Hence $m = m'$. □

4. (Based on Exercise 10.3.)

Solution. ($X \subseteq Y$) Let $x \in X$. Then $x \in \mathbb{Z}$ and $x \equiv -1 \pmod{6}$, so $x + 1 = 6k$ for some $k \in \mathbb{Z}$. Then

$$x - 2 = (x + 1) - 3 = 6k - 3 = 3(2k - 1).$$

Since $2k - 1 \in \mathbb{Z}$, we have $3 \mid (x - 2)$, i.e. $x \equiv 2 \pmod{3}$. So $x \in Y$.

($X \neq Y$) We show $2 \in Y$ but $2 \notin X$. Since $2 - 2 = 0 = 3 \cdot 0$, we have $2 \equiv 2 \pmod{3}$, so $2 \in Y$. But $2 - (-1) = 3$ and $6 \nmid 3$, so $2 \notin X$. Hence $Y \not\subseteq X$ and $X \neq Y$. □